

Predicting Activity Cliffs with Free-Energy Perturbation

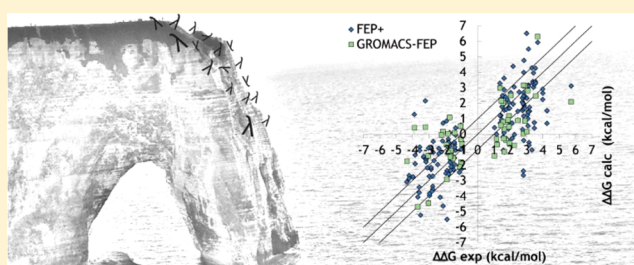
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Supporting Information

ABSTRACT: Activity cliffs (ACs) are an important type of structure–activity relationship in medicinal chemistry where small structural changes result in unexpectedly large differences in biological activity. Being able to predict these changes would have a profound impact on lead optimization of drug candidates. Free-energy perturbation is an ideal tool for predicting relative binding energy differences for small structural modifications, but its performance for ACs is unknown. Here, we show that FEP can on average predict ACs to within 1.39 kcal/mol of experiment (~ 1 log unit of activity). We performed FEP calculations with two different software methods: Schrödinger–Desmond FEP+ and GROMACS implementations. There was qualitative agreement in the results from the two methods, and quantitatively the error for one data set was identical, 1.43 kcal/mol, but FEP+ performed better in the second, with errors of 1.17 versus 1.90 kcal/mol. The results have far-reaching implications, suggesting well-implemented FEP calculations can have a major impact on computational drug design.



INTRODUCTION

Predicting relative protein ligand binding affinities is of high interest for drug discovery.^{1–3} Free-energy simulations provide a rigorous approach. Methods such as free-energy perturbation (FEP) use molecular dynamics (MD) simulations to compute the free-energy difference between two structurally related ligands.⁴ While the theory of FEP dates back several decades,^{5–11} it is experiencing a resurgence due to improved force fields, new sampling algorithms, ease of application, and low-cost parallel computing using graphics processing units (GPU).^{12–14} Modern implementations exploit highly parallelized hardware architecture, such as GPU enabled FEP+ from Schrödinger¹⁵ that uses the Desmond MD-engine.¹⁶ Additional tools have been reported that facilitate the application of FEP in various software.^{17–21} Technological advancements and high interest have led to many interesting applications of FEP and related methodologies.^{22–29} A recent article compared FEP calculated hydration energies with alternative software,³⁰ and the D3R challenge compares different approaches from multiple groups,³¹ but overall there are few reports of different simulation software used for protein–ligand FEP calculations.

Drug discovery lead optimization (LO) involves the synthesis of close analogues around key lead compounds. Activity cliffs (ACs) consist of structurally similar molecules having unexpectedly large differences in bioactivity (generally >2 log units).^{32–34} ACs can have a large impact on an LO program as they represent discontinuities in the SAR

landscape. In early-LO, they can be desirable when offering a departure from a problematic “flat” SAR giving a breakthrough in potency; while in late-LO they can confound the final steps of fine-tuning an advanced lead. Where sufficient data are available, the cause of ACs can be classified as changes in charge, stereochemistry, hydrophobic or polar interactions, water structure, conformational bias, and combinations thereof.³⁵ In principle, FEP should be well suited to capture quantitatively the effect of these changes on compound potency. Previously, we have seen that FEP showed good performance with mean unsigned error (MUE) as compared to experiment <1 kcal/mol for challenging systems with protein flexibility.^{36–40} Being able to predict ACs would be a major advance and allow a systematic computational exploration to identify compound modifications to make or avoid. Therefore, in this work, we apply FEP to study ACs with available structures from the protein data bank (PDB)⁴¹ as well as examples from internal Janssen LO programs.

We report FEP predictions for ACs in 148 related compound pairs (perturbations) across 18 different target proteins, with 33 perturbations performed using both protein X-ray structures solved for each ligand in the AC. Calculations were performed in triplicate ($n = 3$). We used the commercial FEP+ implementation from Schrödinger but also performed

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FEP calculations with GROMACS. This is one of the first studies performing a large-scale comparison of implementations on different MD-engine software. The overall average absolute error on the predicted $\Delta\Delta G$ for all calculated ACs was 1.39 ± 0.07 using the FEP+ approach and 1.65 ± 0.14 kcal/mol for the GROMACS-FEP calculations.

METHODS

Selecting Activity Cliffs. The definition of activity cliffs is based on two criteria, the difference in activity and the similarity of the compounds. The difference in activity was chosen to be at least 100-fold, two log units in pIC_{50} ,³³ although this was relaxed to 10-fold for some ACs within the internal Janssen data sets. Meanwhile, the structural difference was chosen on the basis of matched molecular pairs⁴² with no more than 5 heavy atom differences, as calculated with the MOESaic tool in the MOE software.⁴³ Data sets were drawn from the public and internal domains. All input structures are provided in the [Supporting Information](#).

The public ACs were selected on the basis of crystallized ligands from the PDB and reported bioactivities.⁴¹ We selected 33 AC pairs for 14 different drug discovery relevant protein targets such as Chk2, Chk1, CDK2, BACE1, Thrombin, Renin, etc. In these cases, the protein–ligand complex was solved crystallographically for both compounds, referred to as complex 1 or 2, [Table S1](#). This provided the opportunity to compare the same AC perturbation in two alternative starting structures. The ligands and their binding site interactions are detailed in [Table S2](#). To test an alternative scenario, four Janssen LO programs were datamined to identify the ACs. The aim was to study the performance for predicting ACs within the same chemical series of multiple compounds, an application akin to a drug discovery project. In these cases, at least one ligand was crystallized with the target protein. This gave 115 ACs for four different target proteins and chemical series: BACE1,⁴⁴ PDE2,³⁶ PDE10,⁴⁵ and ROS1⁴⁶ kinase. In most cases, ligand placement for the calculations was trivial. For the PDB data set, ligand input coordinates were prepared by alignment of available crystal structures. In the case of the larger LO data sets, compounds were manually sketched by modification of the crystal structure ligand, ensuring consistent atomic placement of the nonperturbed part of each molecule.

System Preparation FEP+ and Docking. All structures were prepared for computational work using the “Protein Preparation Wizard” tool in Maestro with default settings, missing side chains/atoms were fixed, any missing loops were modeled, protein protonation states were assigned with PROPKA at pH 7.0, metals were retained and zero bond order constraints to neighboring atoms were assigned, the hydrogen-bonding network was optimized, ligand charges assigned, and a brief minimization to root-mean-square deviation (RMSD) 0.5 Å was performed. To compare the FEP-based $\Delta\Delta G$ values with a simple docking scoring function, we also used the same prepared systems to dock the compounds of all ACs with the Glide software.⁴⁷ In the docking calculations, enhanced and expanded sampling were enabled along with postdocking minimization, and all other settings were default. The best predicted ΔG docking score per molecule was used for the $\Delta\Delta G$ AC estimate.

FEP+ Approach. FEP+ calculations were performed using v2018-2 of the Schrödinger modeling suite. The OPLSv3e force field was used, the Desmond (MD) engine v3.9, the REST enhanced sampling technique, and the multistate

Bennett acceptance ratio (MBAR) approach to generate free-energy estimates. In this work, cycle-closure averaging and error estimation could not be used as perturbations were all calculated individually.⁴⁸ The replica exchange with solute tempering (REST) region was applied only to heavy atoms of the ligand. Any missing force field parameters were added by additional QM calculations and fitting using the ffbuilder module. The FEP+ calculations were performed with 12 λ -windows and running 1 and 5 ns production MD simulations per window. Equilibration was performed in five steps: (i) 100 ps at temperature of 10 K with Brownian dynamics, NVT ensemble, solute heavy atom restraints, and small (1 fs) time step; (ii) 12 ps at 10 K with Berendsen thermostat, NVT ensemble, solute heavy atom restraints, and small time step; (iii) 12 ps at 10 K with Berendsen, NPT ensemble, solute heavy atom restraints, increase to default time steps; (iv) 24 ps at 300 K with Berendsen, NPT ensemble, solute heavy atom restraints; and (v) finally, 240 ps at 300 K with Berendsen, NPT ensemble, and no restraints. Production simulations used the NPT ensemble and hydrogen mass repartitioning to permit a 4 fs time step. The BAR analytical error was calculated and is defined as the square root of the estimated variance of the total free energy.^{48–50} Selected perturbations were calculated using the -gcmc flag, which implements a prior solvation step using a Monte Carlo (MC) approach, potentially useful for perturbations that require a difference in water structure between end states. Calculations were performed as three independent repeats using different random seeds, and error bars in the figures are based on their standard deviation of $\Delta\Delta G$. Comparison with experiment is performed via the correlation coefficient (R^2) and the MUE. The definition of activity cliffs implies a “selection bias” because perturbations are chosen from a limited range of $\Delta\Delta G$; this could lead to underestimation of performance. Therefore, we present results across the range of $\Delta\Delta G$, but some caution is needed interpreting R^2 given the grouping at positive and negative limits. MUEs and other mean averages are reported with standard errors. Expected correlation coefficients⁵¹ were calculated by sampling the experimental $\Delta\Delta G$ data 10 000 times with a mean random error, then deriving the overall average correlation coefficient with the original experimental data. Error values of 0.3 log units (0.41 kcal/mol, an approximate experimental error) and 1 log unit (1.37 kcal/mol, the FEP+ MUE) were used.

GROMACS FEP Approach. There are many steps required to set up protein–ligand FEP calculations with GROMACS. A protocol was developed on the basis of several in-house scripts as well as making use of FESetup¹⁹ and alchemical-analysis⁵² tools; see [Figure 1](#). Protein preparation and minimization used antechamber, leap, and sander modules of AMBER.⁵³ The AMBER99SB-ILDN⁵⁴ and general AMBER force field (GAFF)⁵⁵ v2 force fields were used along with the TIP3P water model. FESetup was used to generate the hybrid ligand topologies and the input for only the solvent perturbation leg. Charges were calculated with AM1 bond-charge correction (AM1-BCC) approach. The protein–ligand perturbation input was built using in-house scripts merging protein and the FESetup ligand topologies. In some cases, ligand topologies were modified to correctly account for mapping of atoms in substituents on aromatic rings, noninteger sum of charges, or deletion-insertion consistency in a two-step perturbation. Simulations were submitted and managed via a series of python scripts and run using GROMACS v2016.4.⁵⁶

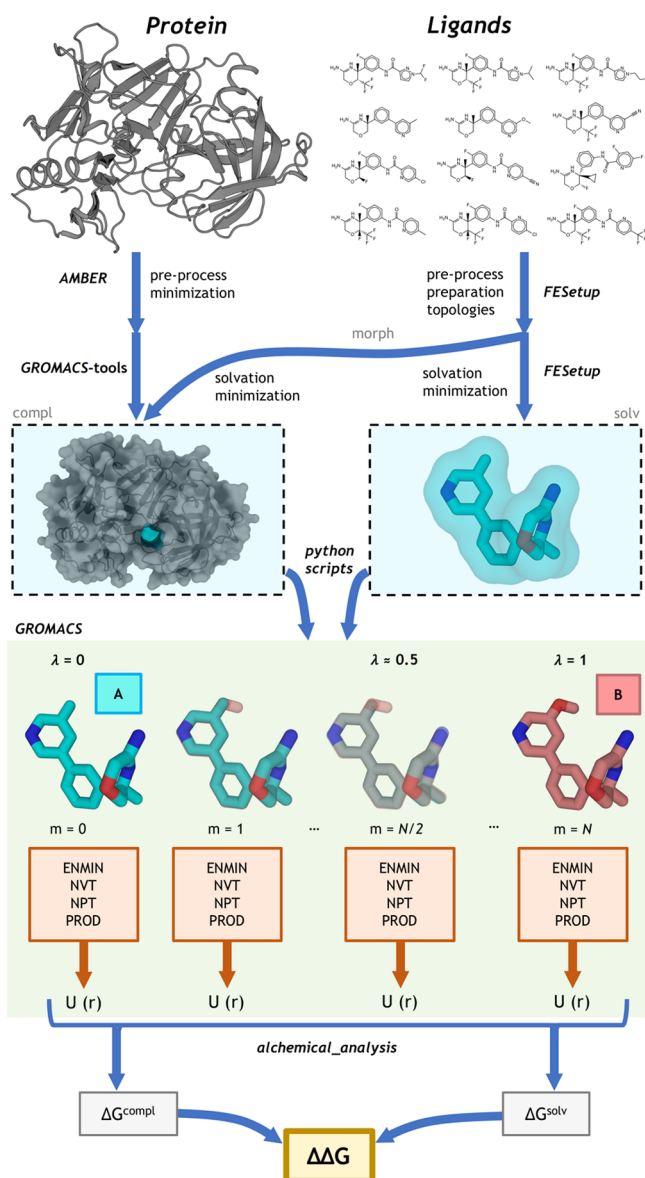


Figure 1. Schematic view of the process for performing FEP calculations with GROMACS. In brief, hybrid ligand topologies were prepared with FESetup, then in-house scripts were used to prepare the protein (with GROMACS-tools) and solvent simulation inputs; merge the ligand and protein topologies (morph step); manage the submission and running of the jobs including minimization, equilibration, and production steps. Alchemical analysis was used to generate the free-energy differences. More details are provided in [Methods](#).

GROMACS calculations used a single topology approach. Perturbations requiring both deletion and insertion proceed via an explicit-intermediate.⁵⁷ Fifteen λ -windows were used with a “split protocol” approach where electrostatics were decoupled during 5 steps then 10 step decoupling of van der Waals and bonded interactions, the order being dependent on whether the perturbation requires deletion or insertion of atoms. The default soft-core potential was used with settings: $sc\text{-}\alpha = 0.5$, $sc\text{-}\sigma = 0.3$, and $sc\text{-}\text{power} = 1$.⁵⁸ Each λ -window included an initial minimization step, a two-step heavy-atom constrained equilibration with NVT (100 ps) and NPT (250 ps) ensembles, followed by 1 ns unconstrained production MD using NPT. The simulation settings included using a 2 fs time

step with the sd integrator, a temperature of 300 K, Parrinello–Rahman barostat, waters constrained with LINCS, the Particle Mesh Ewald (PME) method to calculate electrostatic interactions, and the Verlet scheme for neighbor cut-offs. The free-energy differences were generated using the MBAR approach with the python package alchemical-analysis. Input structures and example GROMACS input files are provided in the [Supporting Information](#).

RESULTS

FEP+ Calculations of 3D Activity Cliffs. There were 33 AC pairs in the public 3D data set, and each protein–ligand complex was solved crystallographically for both ligands in each AC, referred to as complex 1 or 2. [Figure 2](#) compares

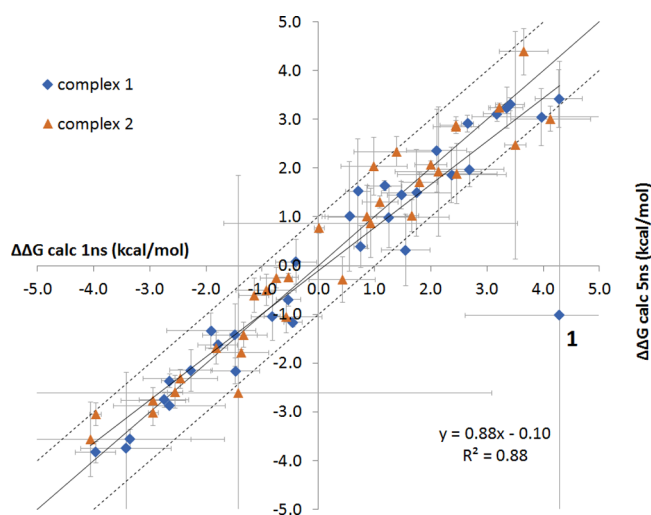


Figure 2. Comparing 1 and 5 ns sampling time per λ -window for the FEP+ calculation of $\Delta\Delta G$ for 3D activity cliffs. Calculations were performed using either complex 1 or 2 as the input protein structure. All data are the average of three repeats, and error bars show standard deviation of repeats. Numbering, 1 in this case, identifies outliers for later discussion.

calculations performed with 1 and 5 ns λ -windows and with three repeats for all 66 $\Delta\Delta G$'s. Even for this structurally diverse data set, there was a high correlation, R^2 0.88, between the 1 and 5 ns λ -window simulation times; also, the slope and intercept were close to 1 and 0, respectively. Altogether this suggests little difference with a 5-fold increase in sampling. The mean standard deviation of repeats for 1 and 5 ns was similar, 0.58 ± 0.08 and 0.56 ± 0.11 kcal/mol (median 0.42 and 0.32), respectively. Regarding the variability of calculated results with repeat calculations, 7 of the 66 perturbations performed with the 1 ns protocol and 6 with the 5 ns protocol showed a standard deviation >1 kcal/mol. The same perturbation involving Leukotriene A4 hydrolase had a standard deviation of 4.50 and 5.21 kcal/mol for 1 and 5 ns protocols, respectively. This perturbation was the single major outlier 1 in the bottom right quadrant of [Figure 2](#). The mean average Bennett error over all individual perturbations was 0.22 kcal/mol, with minimum, maximum, and standard deviation of 0.04, 0.92, and 0.13 kcal/mol, respectively. The Bennett error and the standard deviation of repeats were correlated, R^2 between 0.42 and 0.68 for the different length λ simulation and input complexes. For low Bennett errors (<0.2 kcal/mol), the standard deviation of repeats was also low; only 0.5% of

examples showed a standard deviation >1 kcal/mol. Whereas for Bennet error >0.4 kcal/mol, 78% of examples showed a standard deviation >1 kcal/mol. Therefore, the Bennet error can be informative for the resulting standard deviation from repeat experiments.

Before comparing with experiment, we consider the $\Delta\Delta G$ calculated for the same perturbations but using the alternative complex 1 or 2 crystal structures. Results for the two input structures became more similar as simulation time was increased: correlation R^2 of 0.33 and 0.51 for 1 and 5 ns λ -window simulations, respectively (Figure 3). The average

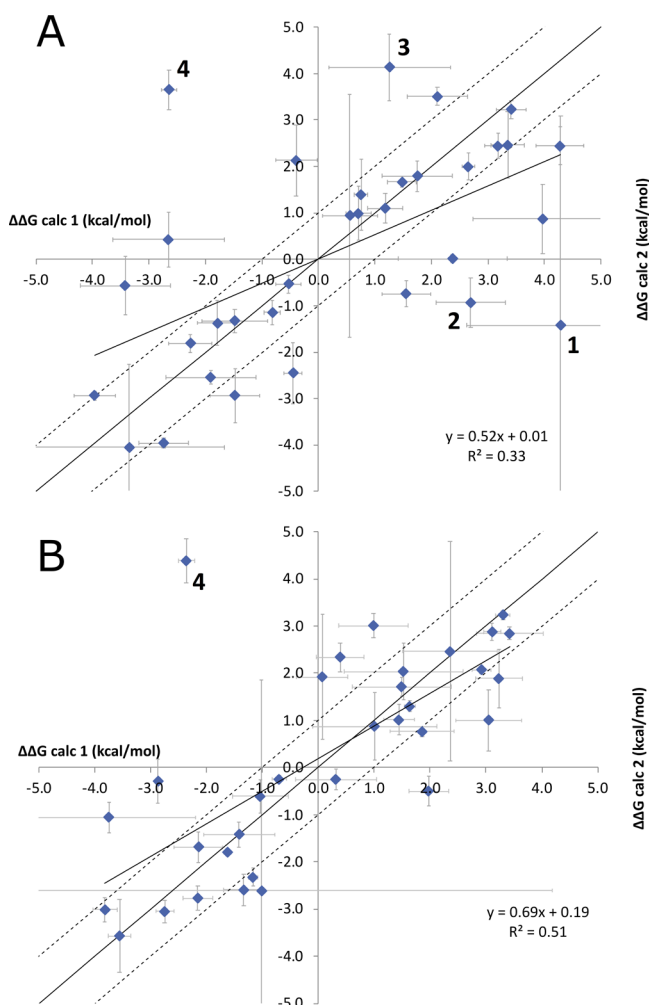


Figure 3. Comparing the FEP+ calculation of $\Delta\Delta G$ for 3D activity cliffs when using either complex 1 ($\Delta\Delta G$ calc 1) or complex 2 ($\Delta\Delta G$ calc 2) as the starting protein structure. Calculations were performed using a 1 ns λ -window protocol (A) or 5 ns (B). All data are the average of three repeats, and error bars show standard deviation of repeats. Numbering identifies outliers for later discussion.

absolute difference reduced accordingly from 1.53 ± 0.27 to 1.09 ± 0.23 kcal/mol for 1 and 5 ns. For 21 out of the 33 perturbations, the calculated $\Delta\Delta G$ for complexes 1 and 2 was more similar with the 5 ns as compared to the 1 ns protocol. While the overall 1 and 5 ns results were highly correlated (Figure 2), it is understandable that increased sampling leads to more similar “converged” calculated $\Delta\Delta G$ ’s when using different starting structures. The average, minimum, and maximum heavy-atom RMSD between complex 1 and 2 input structures were 1.1, 0.2, and 2.8 Å, respectively. Among

the 21 perturbations that had more similar $\Delta\Delta G$ values with increased simulation time, for 7 of these the calculated $\Delta\Delta G$ with 5 ns was at least 1 kcal/mol more similar than with 1 ns simulations. These 7 perturbations had an average RMSD difference between complex 1 and 2 of 1.3 Å, slightly larger than the average for the total set (1.1 Å), confirming that those that improved the most were on average those that had the more different structures. The trend was clearer when comparing the binding pocket amino acids. The average RMSD for the 7 perturbations that delivered more similar results with increased sampling had an average RMSD in their initial states of 3.8 Å as compared to 1.3 Å for the total set. Several perturbations among the 7 involved a DFG-out/loop-in versus DFG-in/loop-out conformational change for the two similar ligands bound to Aurora-A kinase, for example, 3UOH-4DEA in Table S2. Overall, the extra sampling reduced the impact of varying input structures on the $\Delta\Delta G$, and roughly two-thirds of the perturbations showed more consistent behavior with longer sampling. Other case specific factors such as local solvent or side chain movements may also be at play and still not fully resolved with the 1 to 5 ns per λ -window sampling increase, requiring even longer λ -windows to converge.

Figure 4 shows the calculated $\Delta\Delta G$ versus experiment for each of the 33 perturbations performed using the complex 1 protein structure. As discussed (Figure 2), the results for 1 and 5 ns λ -window protocols were similar. The average MUE between calculated and experimental values was close to one log unit in activity, 1.37 ± 0.18 and 1.46 ± 0.19 kcal/mol, respectively, for the two simulation protocols (Table 1). The average experimental change in activity for the set of 33 ACs was -3.24 ± 0.08 kcal/mol, but the average calculated value was only -2.11 ± 0.22 kcal/mol. The MUEs between FEP+ predicted $\Delta\Delta G$ and experiment were all significantly better than the predicted changes in activity arising from docking, where the MUE was higher, from 2.91 ± 0.18 to 3.83 ± 0.25 kcal/mol. As mentioned above, the calculations were performed using two starting protein conformations corresponding to the X-ray structures of both the high and the low affinity molecules of the AC. Choosing the $\Delta\Delta G$ with the best agreement with experiment in each case, resulted in an overall MUE of 1.03 ± 0.16 and 1.12 ± 0.18 kcal/mol for 1 and 5 ns λ -window protocols, respectively. Averaging the results for the two complex structures led to a slightly worse MUE of 1.45 ± 0.17 and 1.56 ± 0.18 kcal/mol for 1 and 5 ns λ -window protocols, respectively. This was due to the detrimental effect of some poorly predicted $\Delta\Delta G$ ’s on the average MUE.

Returning to the complex 1 results, the R^2 correlations between the calculation and experiment were 0.64 and 0.65 for the 1 and 5 ns simulation protocols, respectively. The expected R^2 values were 0.98 and 0.79 based on either experimental or FEP errors, respectively. As expected, the calculated R^2 did not outperform the R^2 based on experimental error and came close to the R^2 assuming FEP error, suggesting good performance for the data sets. Some caution is needed as the high R^2 values reflect the clustering of data at each end of the experimental range. Nevertheless, docking did not show any correlation with experiment. The two plots reveal a similar number of compounds (13 and 14 for Figure 4A and B, respectively) with a calculated $\Delta\Delta G$ within 1 kcal/mol of experiment, and the majority of remaining perturbations were underestimated versus experiment. Indeed, considering all 132 calculations, using complex 1 and 2 as input geometries and 1 and 5 ns λ -

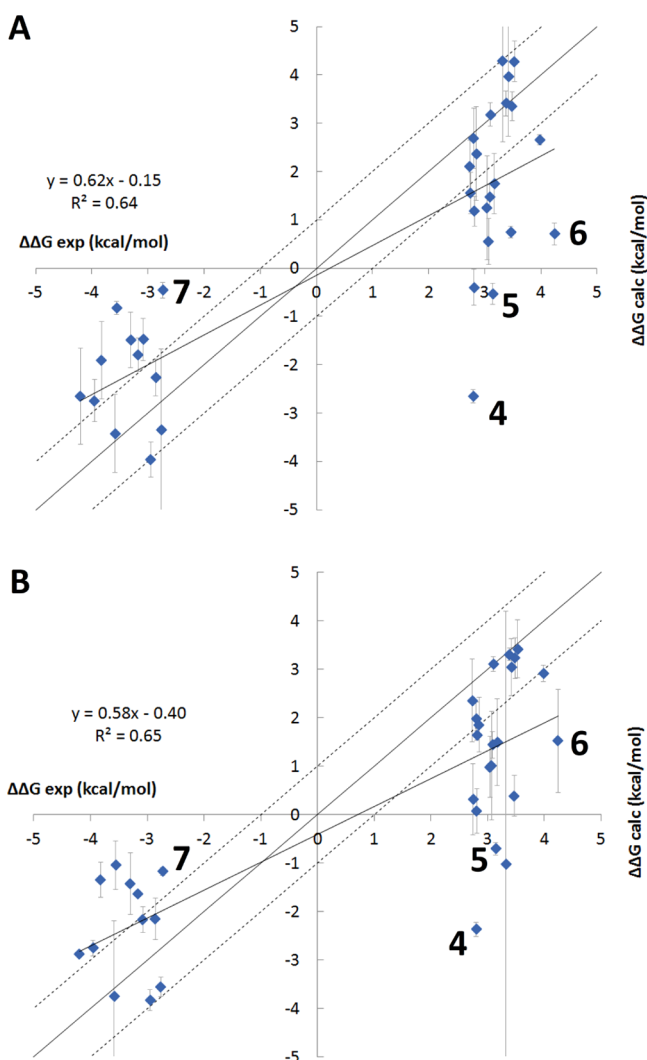


Figure 4. Comparing the FEP+ calculated $\Delta\Delta G$ with experiment for 3D activity cliffs. Performed with complex 1 starting protein structure and calculations used a 1 ns λ -window protocol (A) or 5 ns (B). All data are the average of three repeats, and error bars show standard deviation of repeats. Numbering identifies outliers for later discussion.

window sampling, only three calculated $\Delta\Delta G$'s were over-predicted as compared to experiment by 1 kcal/mol or more. Finally, the majority of $\Delta\Delta G$ predictions had a correctly predicted sign for the change in activity, 90% (119 out of 132), whereas for docking this reduced to just 52%.

FEP+ Calculations on LO-Series Activity Cliffs. Internal data sets were mined using matched molecular pairs analysis to identify ACs, and the FEP+ predicted versus experimental $\Delta\Delta G$ are shown in Figure 5. Most of the perturbations had a correctly predicted sign for the change in activity, 87% (100 out of 115), consistent with 89% (219 out of 247) for all FEP+ calculations in this article. The PDE2, PDE10, and ROS1 data sets performed best in terms of R^2 correlation: 0.78, 0.70, and 0.85, respectively, while the BACE1 data set showed worse correlation, R^2 0.39. As expected, none of the calculated results outperformed the R^2 with experimental error: the expected R^2 assuming a relatively small experimental error (0.41 kcal/mol) were all very high, ≥ 0.95 . For three of the four data sets, the calculations showed better correlation than the expected R^2 based on a FEP error of 1.37 kcal/mol. This was simply a reflection of the lower MUEs in the range 0.77–1.43 kcal/mol than that used for the expected R^2 estimation. In each case, the MUE was significantly better than docking with p values ranging from <0.05 to <0.001 depending on the data set (Table 1). For BACE1, a series of perturbations corresponding to a similar type of chemical change were incorrectly predicted to be unfavorable as compared to experiment, outlier 8 (Figure 5). Meanwhile, for PDE10 a subset was overpredicted leading to a slightly increased slope (1.11x).

FEP GROMACS Calculations on LO-Series Activity Cliffs. GROMACS-FEP was performed to predict the $\Delta\Delta G$ for BACE1 and PDE2 ACs, and the results are compared to experiment and FEP+ in Figure 6. For the BACE1 and PDE2 data sets, 73% and 64% of the perturbations had a correctly predicted sign for the change in activity. This was compared to 70% and 88% for the calculations on the same data sets with FEP+. For the BACE1 data set, the R^2 correlation with experiment was 0.47 and the MUE was 1.44 ± 0.20 kcal/mol, which was comparable with FEP+, R^2 0.39, and MUE 1.43 ± 0.25 kcal/mol (Table 1). This was confirmed by inspecting the correlation between results from the two simulation software (Figure 6) that showed a relatively high correlation, R^2 0.83.

Table 1. Comparison of Calculated $\Delta\Delta G$ Values with Experiment for Activity Cliffs from the Different Data Sets Studied

	<i>n</i>	FEP+		docking		expected R^{2e}	
		R^2	MUE ^a	R^2	MUE ^a	expt. error	FEP error
3D PDB ACs							
complex 1, 1 ns/ λ	33	0.64	1.37 ± 0.18^d	0.00	3.83 ± 0.25	0.98	0.79
complex 1, 5 ns/ λ	33	0.65	1.46 ± 0.19^d				
complex 2, 1 ns/ λ	33	0.56	1.79 ± 0.23^d	0.01	2.91 ± 0.18		
complex 2, 5 ns/ λ	33	0.60	1.68 ± 0.22^d				
best 1 ns/ λ result ^f	33	0.74	1.03 ± 0.16^d	0.00	2.57 ± 0.20		
best 5 ns/ λ result ^f	33	0.74	1.12 ± 0.18^d				
LO Data Sets							
BACE1	30	0.39	1.43 ± 0.25^b	0.07	2.42 ± 0.31	0.96	0.66
PDE2	26	0.78	1.17 ± 0.15^d	0.00	2.37 ± 0.23	0.97	0.72
PDE10	33	0.70	1.26 ± 0.16^c	0.00	2.24 ± 0.26	0.95	0.62
ROS1	26	0.85	0.77 ± 0.09^b	0.42	1.87 ± 0.20	0.95	0.61

^akcal/mol and all $\pm$ standard error. ^b $p < 0.05$. ^c $p < 0.01$. ^d $p < 0.001$; p -value when comparing the mean MUE for docking to FEP+ using two tailed t test. ^eBased on assuming an experimental or FEP error of 0.41 or 1.37 kcal/mol, respectively. ^f“Best” refers to taking the complex 1 or 2 result with the lowest MUE for each perturbation studied.

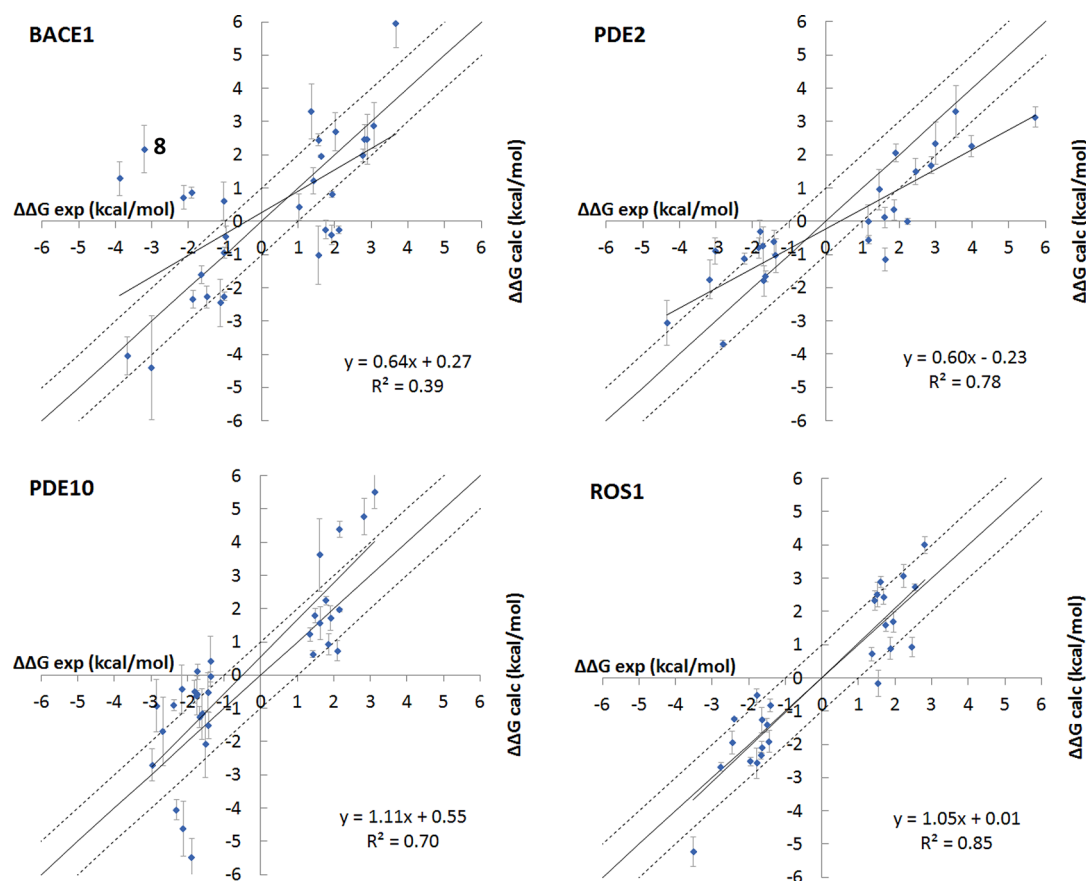


Figure 5. Comparing the FEP+ calculated $\Delta\Delta G$ with experiment for LO-series activity cliffs. Calculations used a 1 ns λ -window protocol, each data point represents a perturbation between two molecules, the calculated value is the average of three repeats, and error bars show the standard deviation. Numbering identifies outliers for later discussion.

There were common outliers such as 8 for the two software protocols. As mentioned above, these correspond to a similar type of chemical change that was incorrectly predicted to be unfavorable as compared to experiment. Meanwhile, for the PDE2 data set, the R^2 correlation with experiment was 0.44 and the MUE was 1.90 ± 0.20 kcal/mol. This was worse than the FEP+ application, which showed R^2 of 0.78 and MUE of 1.17 ± 0.23 kcal/mol. Despite this, there was reasonable correlation between the two methods with an R^2 of 0.66 (Figure 6). The correlation reveals no major outliers; instead the difference in MUE arises from the narrower $\Delta\Delta G$ range of GROMACS-FEP as compared to FEP+.

DISCUSSION AND CONCLUSIONS

We have presented a large test of the performance of FEP for calculating ACs. Two FEP implementations were used for the AC predictions: FEP+ and GROMACS. Docking was used as a relatively simple comparator. The FEP+ results were similar for 1 and 5 ns λ -window simulations. In some cases, repeats showed large standard deviations. Using different starting complex protein structures also led to varied results. The agreement with experiment suggested the clear majority of predicted $\Delta\Delta G$'s had the correct sign, and that performance was better than docking; the overall average error as compared to experiment was ~ 1.4 kcal/mol. The GROMACS method revealed qualitatively similar results, suggesting major success or failure is a function of system setup rather than the differences in the underlying methods. However, the PDE2

data set revealed greater accuracy for FEP+ and a better range of predicted $\Delta\Delta G$. In the following sections, the topics arising from this study are discussed.

Simulation Time. In this work, there was little variation in comparison with experiment arising from 1 or 5 ns λ -windows. When considered along with our previous reports, it becomes clear that improvements due to increased sampling are not universal. We have seen improvements for a few outliers but never an entire data set,³⁷ or for difficult large perturbations,³⁸ but in this study little difference was identified. For outliers in the LO BACE1 data set (Figure 5, top-left points), we increased the λ -window simulations to 20 ns, but the average MUE as compared to experiment worsened to 5.88 as compared to 5.27 kcal/mol for the same perturbations with 1 ns λ -windows. Nevertheless, increased sampling did improve the consistency between results arising from alternative starting structures.

Differences in Protein Structure. Differences in results were seen when performing calculations using two alternative protein structures. As mentioned, increased sampling led to increased correlation between the $\Delta\Delta G$ prediction calculated using complexes 1 and 2. Complex 1 performed better for $\Delta\Delta G$ prediction than complex 2, Table 1. In 20 out of the 33 perturbations, complex 1 was the higher affinity complex of the AC. Also, the average MUEs for the $\Delta\Delta G$ when calculated in either the high or the low affinity complex starting structures were 1.37 ± 0.19 and 1.79 ± 0.21 kcal/mol, suggesting the higher affinity structures led to lower errors (p 0.07). There were no significant differences in structural quality criteria

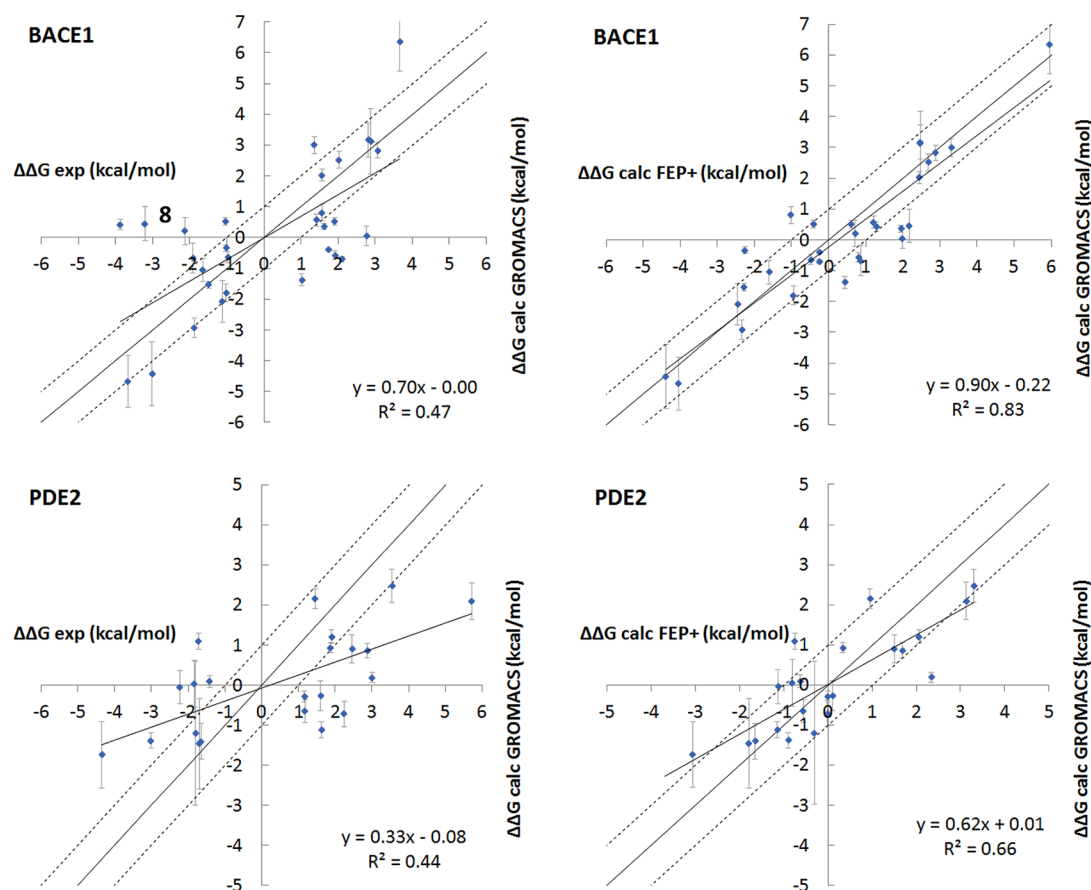


Figure 6. Comparing the GROMACS-FEP calculated $\Delta\Delta G$ with experiment (left) and with FEP+ calculated $\Delta\Delta G$ (right), for BACE LO-series activity cliffs. Calculations used a 1 ns λ -window protocol, each data point represents a perturbation between two molecules, the calculated value is the average of three repeats, and error bars show the standard deviation. Numbering identifies outliers for later discussion.

between the high and low affinity complexes: the average resolutions were 2.11 and 2.10 Å, average R -free were 0.24 and 0.24, and the average isotropic B -factors were 33.0 and 33.8 Å², for each set of high and low affinity structures, respectively. The challenge for ACs is to accurately estimate the magnitude of change in activity (the sign was correctly predicted in 89% of cases). It seems likely the better accuracy of the high affinity complexes arises from the experimental representation of small, case-specific, differences (such as water structure) that may be crucial for accurate energy calculations but hard to predict when starting from the low affinity complex.

Outliers. Selected outliers were numbered in Figures 2–6, and their chemical structures and binding site environments are summarized in Figure 7. Outlier 1 showed a substantial difference in predicted $\Delta\Delta G$ between 1 and 5 ns λ -window simulation time (Figure 2); it also had an extremely large standard deviation. This perturbation involved a relatively small change in 2D chemical structure, with no change in formal charge as both species remain zwitterionic (Figure 7). However, the amino acid group reorients, and the amine occupies a different subpocket changing the metal coordination. Hence, this is a more dramatic structural change than expected and involves a changing metal interaction that may not be well captured by the force field.

The same perturbation was an outlier in the correlation of complex 1 and 2 results (Figure 3) along with examples 2–4. For outlier 2, the change in substitution position of the hydroxyl group led to an alternative conformation of the pyran

ring and water displacement, causing the difference in using the two alternative crystal structures. Water displacement was again the origin of the difference in $\Delta\Delta G$ calculations for outliers 3 and 4. In the former, the azepine ring was puckered upward, displacing a water and interacting directly with Asp267, while the piperidine was more planar with respect to the fused benzyl ring, and interacted with Asp267 via a water bridge. For outlier 4, the simple perturbation, modification of hydrogen for chlorine, also led to water displacement.

Various outliers were identified in the comparison with experiment (Figures 4 and 5). Mentioned above, 4 was also an outlier with respect to experiment, while outlier 5 was another relatively straightforward ligand perturbation but with inaccuracies due to water displacement. Outlier 6 was more complicated. The aminoguanidine group formed multiple hydrogen-bond interactions as well as a salt bridge with Glu273. The structural perturbation of the ligand was trivial, modification of hydrogen to hydroxyl. Both ligands were modeled with a single positive formal charge. However, the change of H to OH modifies the basicity and electronics of the guanidine, and the result could be a complex scenario of protomeric and tautomeric states. Other outliers (not labeled) in the vicinity of 6 in Figure 4A were similar Chk2 perturbations. Outlier 7 was an example that improved with longer simulation time. It involved small perturbations in three different regions of the ligand, and longer time was needed for convergence and adaption of interactions (such as pyridyl

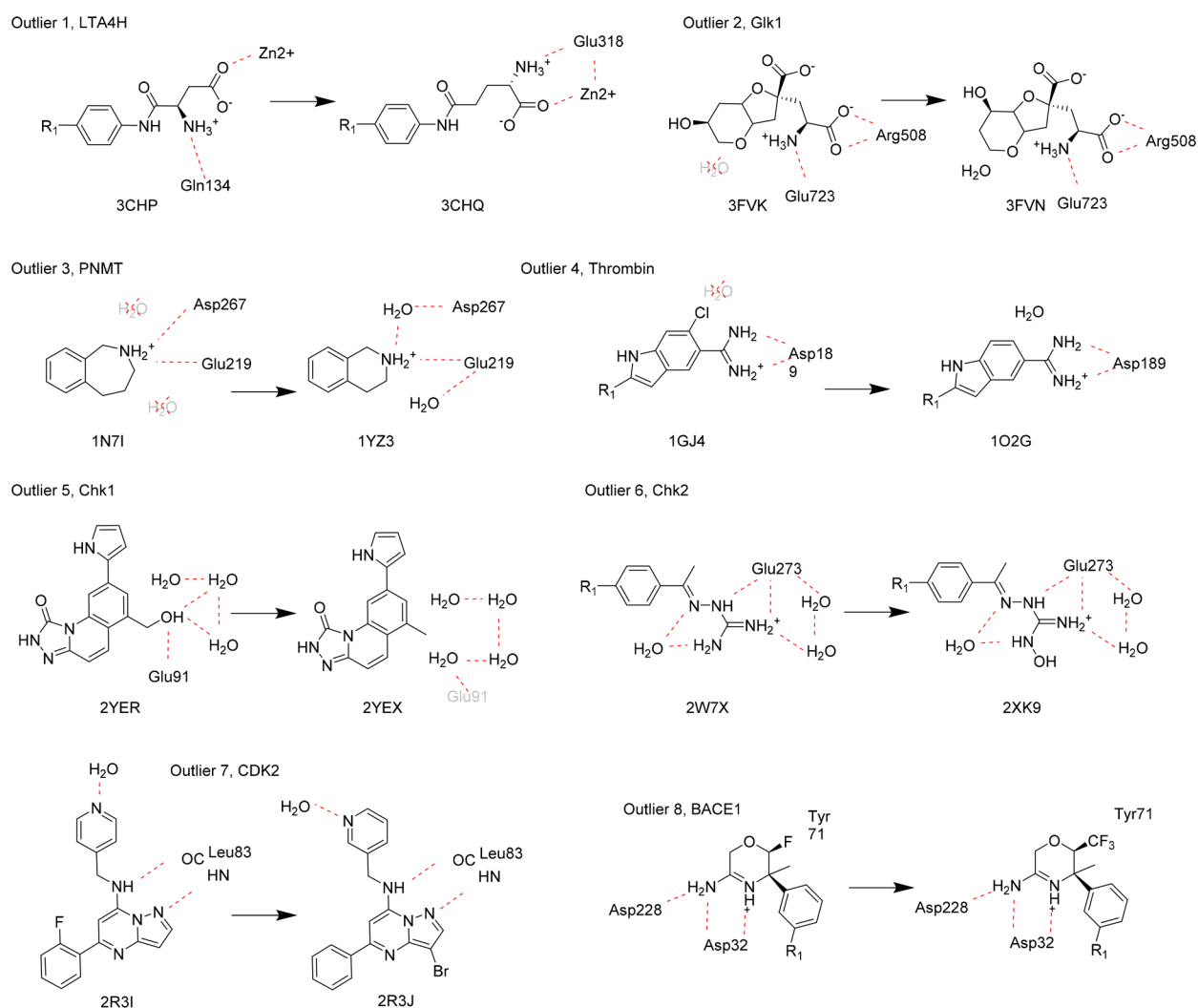


Figure 7. Illustration of selected outliers identified by numbering in Figures 2–6.

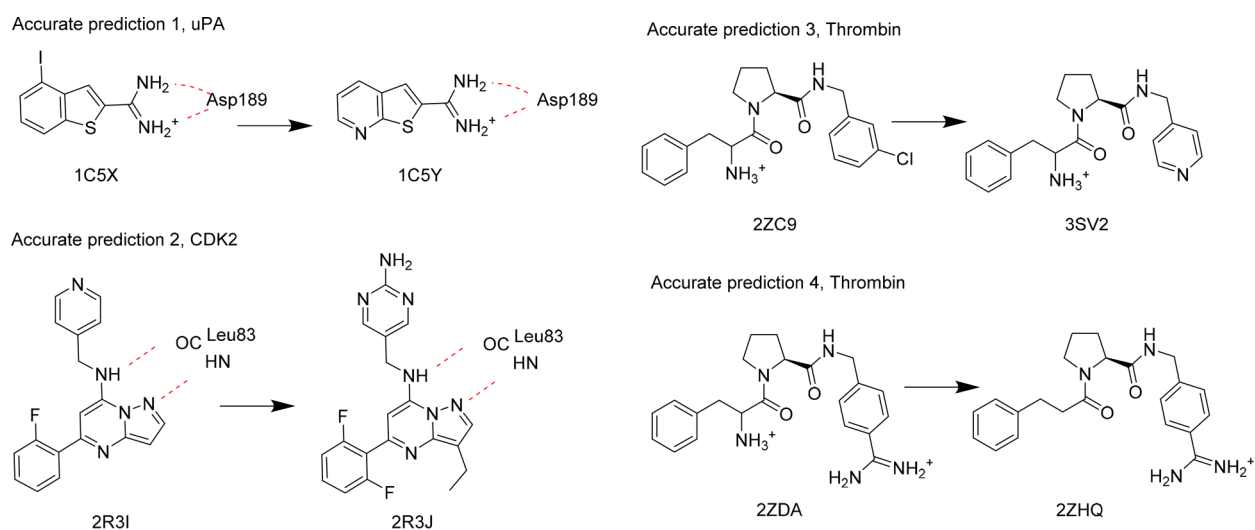


Figure 8. Examples of selected perturbations that performed well as compared to experiment.

nitrogen to water). Finally, **8** was an FEP+ and GROMACS-FEP outlier as compared to experiment (Figures 5 and 6). The BACE1 inhibitors involve modification of a more potent CF₃ to a single fluorine (Figure 7) on a partially saturated cyclic

amidine. We have previously suggested this is due to a pseudoanomeric effect,⁴⁴ and it seems here to present a challenge for the FEP methods. The phenyl requires an axial orientation for optimal BACE1 inhibition; however, the single

F substituent prefers to be axial, thereby shifting the Ph group to equatorial. In contrast, the CF₃ adopts a favorable equatorial conformation permitting the Ph to be axial as needed. This was the cause of additional BACE outliers in the vicinity of **8** in Figure 5, all involving F to CF₃, F to CHF₂, F to Me, and F to OMe at the same position. Several outliers resulted from the difficulty to model water displacement. This is a known limitation for FEP due to the need for the same input configuration for both end states. Performing a prior solvation step with a MC method to solvate buried pockets for some states, and then correcting that all λ windows have the same number of atoms, can overcome this problem. We applied this method to FEP+ calculations of selected outliers. For 1 ns $n = 3$ perturbations, the MUE compared to experiment both with and without using the presolvation was 2.61 versus 5.45, 3.83 versus 3.67, and 0.28 versus 2.28 kcal/mol for outliers **4**, **5**, and **7**, respectively. Thus, for outlier **5**, there was no change, whereas outliers **4** and **7** showed improvement, suggesting the value of an MC presolvation step for such cases.

Although we have focused on outliers because of their interest for future work, many activity cliffs were accurately predicted as compared to experiment, Figure 8. These included small structural perturbations expected to behave well (PDBs 1CSX to 1CSY, and 2ZC9 to 3SV2), but also modification in several positions in the same ligand (PDBs 2R3I to 2R3J) and even charge change (PDBs 2ZDA to 2ZHQ), a capability that is enabled in the FEP+ approach.

FEP+ versus GROMACS-FEP. The implementation of FE calculations can vary across software depending on the availability of algorithms for integration, temperature and pressure scaling, electrostatic and van der Waals cut-offs, and even the precision of physical constants;⁵⁹ see Loeffler et al.³⁰ for an excellent discussion. Beyond those details, our calculations have several even more obvious differences: inclusion or not of REST, AMBER+GAFF versus OPLS3e force fields, single versus dual topology, etc. Therefore, despite these differences, the resulting similarity between FEP+ and GROMACS-FEP for the BACE1 and PDE2 LO data sets was surprising. This indicates that a substantial part of the accuracy of FEP $\Delta\Delta G$ calculations is fundamental to the system setup, starting conformation, protein and/or ligands preparation, etc. The GROMACS-FEP results for the BACE1 data set were comparable with those of FEP+, with similar outliers and high correlation. For the PDE2 data set, the FEP+ method performed better as GROMACS-FEP had a smaller range of $\Delta\Delta G$.

We performed calculations using GROMACS-FEP on other systems in this study. In general, the initial “out-of-the-box” performance revealed lack of FEP convergence arising from poor λ -overlap. One common cause was inconsistent movements and drifting of one ligand with respect to the other. When performing FEP, we are usually concerned with inadequate sampling, but we also see issues due to inappropriate sampling. It became clear that a more bespoke treatment of each system would likely be needed. This was not surprising as expert application of MD-based methods can require significant investment of time to develop ideal simulation setup and parameters, as with GPCRs for instance. The next steps could include testing stability via classical MD simulations, running FEP from classical MD snapshots, very large numbers of replicates,⁶⁰ testing alternative force fields for the entire system,⁶¹ combining force fields,⁶² or detailed ligand charge and parameter modifications. None of these could be

applied today with reasonable throughput over many systems and ligand perturbations. Overall, our results show that GROMACS-FEP can deliver accurate results, but many aspects of system setup require deep validation.

Value of Repeats. In this study, all calculations were performed as 3 repeats with alternative random seeds and results presented as the mean with standard deviations. The average standard deviation for all 1 ns λ -window perturbations was 0.39 kcal/mol for FEP+ and 0.38 kcal/mol for GROMACS-FEP calculations, remarkably similar. As can be seen in the various correlation plots, some perturbations had high variation, with the largest standard deviation being 1.56 kcal/mol. As seen elsewhere,^{60,63,64} repeat calculations are likely essential for some systems, and predictions may improve with more than triplicate repeats used here. The results generally underestimate the change in activity as compared to experiment. Errors in FEP originate from either force field, sampling, or inappropriate treatment of the system.⁶⁵ Here, the trend to underestimate is likely due to the inability to sample the energy barriers involving the change in affinity of the activity cliff.⁵⁷ This was not resolved by increasing sampling by an order of magnitude, for instance 1, 5, or 25 ns as tried here; instead it may be advantageous to cover more conformational space via many repeats as compared to one or only three single long trajectories.⁶⁴

In this work, the $\Delta\Delta G$'s were all calculated directly as individual perturbations. Typically, in a drug discovery setting, FEP calculations would be performed on sets of compounds with each compound involved in several perturbations with close analogues. The resulting $\Delta\Delta G$'s are then corrected on the basis of cycle closure averaging and weighting rules. This can lead to beneficial effects akin to performing repeats and averaging.⁴⁸ Hence, the individual perturbations performed here likely represent the most challenging use case scenario.

Computational Cost. Our work used two alternative approaches. While the final results for completed data sets were qualitatively similar, it is also worthwhile to consider the different computational cost. The two softwares rely upon different hardware architectures: FEP+ Desmond entirely on GPU, while GROMACS-FEP runs on a mixed CPU-GPU node. A 1 ns 12 λ -window FEP+ perturbation on the LO BACE1 system required an average of 34 and 86 min for solvent and complex, respectively. These times include all 12 λ -windows run in parallel on the same 4 GPUs (Nvidia Tesla K80). The GROMACS-FEP calculation of the same BACE1 solvent and complex perturbations required approximately 30 and 180 min for each λ -window on 16 CPU cores and 2 GPU (Nvidia Tesla K40). For this hardware, the difference is ~12–25-fold increased performance for FEP+. However, these differences reduce if one equates the number and performance of the GPUs (2 $\times$ number of GPUs, and approximately 2 $\times$ performance of K80 vs K40) to a 3–6-fold improvement.

FEP approaches and technology are advancing,⁶⁵ and new data sets are needed to continue improving FE prediction methods.^{66,67} We have presented a large study of FEP involving ACs from relevant drug discovery targets. We conclude that the changes in activity could be predicted with an accuracy of ~1.4 kcal/mol, over the whole data set. Best case scenarios in the literature suggest favorable LO drug discovery data sets can be predicted with MUE < 1 kcal/mol. Thus, it seems that certain aspects of ACs provide a challenge for current FEP, such as metal ions, water displacement, and possible tautomeric effects. Nevertheless, the results were

encouraging and would suggest FEP could be used to rank ACs among other perturbations especially given their inherent large change in experimental activity. Finally, we demonstrated a relatively large application of GROMACS-FEP, to our knowledge, the largest protein–ligand study with this popular MD engine. Similar results were seen with FEP+ for one data set, while for the second FEP+ showed better accuracy. There is ample literature evidence to suggest the origins of this improvement within the FEP+ technology,¹⁴ but, nevertheless, we consider the results reassuring and positive for the field. Given the interest in the literature, one could consider how data-driven machine learning techniques may perform at this task. Supposing similar perturbations exist in the prior art, then machine learning approaches may perform adequately, particularly new 3D structure-based machine learning approaches. Otherwise, activity cliffs are a consequence of a change in multiple 3D parameters in tandem: such as ligand, protein, plus water conformation, thus difficult to capture without resort to physics-based modeling.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.8b01290.

Details of the 3D activity cliffs and calculation input structures (PDF)

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Notes

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■ ABBREVIATIONS

AC, activity cliff; BACE1, beta-secretase 1; FEP, free-energy perturbation; GAFF, general amber force field; LO, lead optimization; MD, molecular dynamics; MUE, mean unsigned error; ns, nanoseconds; PDB, protein data bank; PDE2, phosphodiesterase 2; PDE10, phosphodiesterase 10; REST,

replica exchange with solute tempering; RMSD, root-mean-square deviation; ROS1, C-ros oncogene 1 receptor tyrosine kinase.

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